

Processando config 7, seed 0, teste: Rabbit
XX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

smultin_MultinomialBitsOver test:

N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = $1 / 32768$
EColl = $n^2 / (2k)$ = 128
Hashing = TRUE

Collision test
CollisionOver: density = $n / k = 1 / 32768.00$
Expected number of collisions = $\mu = 128.00$

Results of CollisionOver test:

POISSON approximation :
Expected number of collisions = $N \cdot \mu$: 256.00
Observed number of collisions : 232
p-value of test : 0.93

Total number of cells containing j balls

j = 0	:	549739036904
j = 1	:	16776752
j = 2	:	232
j = 3	:	0
j = 4	:	0
j = 5	:	0

CPU time used : 00:00:04.43

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 18.00
p-value of test : 0.12

CPU time used : 00:00:00.06

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.60
p-value of test : 0.28

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 8.08
p-value of test : 0.33

Normal statistic for number of jumps : 0.62
p-value of test : 0.27

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.48
p-value of test : 0.32

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.99
p-value of test : 0.16

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.055
p-value of test : 0.04

Kolmogorov-Smirnov- statistic = D- : 0.022
p-value of test : 0.61

Anderson-Darling statistic = A2 : 1.68
p-value of test : 0.14

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 0.95
p-value of test : 0.81

Global longest run of 1 : 24.00
p-value of test : 0.39

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 104.98
p-value of test : 0.21

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 20.34
p-value of test : 0.56

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.98
p-value of test : 0.84

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.70
p-value of test : 0.24

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.83
p-value of test : 0.20

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 186.18
p-value of test : 0.30

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 299.26
p-value of test : 0.58

CPU time used : 00:00:00.02

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 460.21
p-value of test : 0.82

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :    0.39
p-value of test          :    0.35
-----
```

```
CPU time used            : 00:00:00.02
```

```
Generator state:
269499904 bits have been read.
```

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*****
```

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HOST = matheus-IdeaPad-3-15ALC6, Linux
```

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :    1.64
p-value of test          :    0.05
-----
```

```
CPU time used            : 00:00:00.00
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin
```

```
sstring_Run test:
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```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355443
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 30.81
p-value of test : 0.71
-----
```

```
-----
Total number of bits: 13416896
```

```
Normal statistic for number of bits : -0.94
p-value of test : 0.83
```

CPU time used : 00:00:00.05

Generator state:
299693984 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 2.49
p-value of test : 0.48

CPU time used : 00:00:00.06

Generator state:
316471200 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.08

Generator state:
333162400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	39.64
p-value of test	:	0.58

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	36.70
p-value of test	:	0.70

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	71.52
p-value of test	:	0.24

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	31.88
p-value of test	:	0.66

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	25.81
p-value of test	:	0.26

CPU time used : 00:00:00.05

Generator state:
349939616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	95.32
p-value of test	:	0.44

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	101.73
p-value of test	:	0.28

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	480.45
p-value of test	:	0.84

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	108.28
p-value of test	:	0.09

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	37.82
p-value of test	:	0.90

CPU time used : 00:00:00.05

Generator state:

366716832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config7_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	141.18
p-value of test	:	0.24

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	154.66
p-value of test	:	0.08

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	169.33
p-value of test	:	0.31

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	161.36
p-value of test	:	0.03

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	95.82
p-value of test	:	0.29

CPU time used : 00:00:00.03

Generator state:

383493632 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.06
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
-----	-----

All other tests were passed